

Ayman Charania

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EDUCATION

- Rice University** Houston, TX
• *BA in Computer Science; GPA: 4.0 (President's Honor Roll)* Aug. 2022 – May 2026
Courses: Computational & Algorithmic Thinking, Computer Engineering, Linear Algebra, Functional Programming, Concurrent Program Design

EXPERIENCE

- Software Engineering Intern** Beaumont, TX
• *Xplor Spot Munro's DCC — JavaScript, HTML, CSS, SQL, Node.js* May 2024 - August 2024
 - **Backend System Enhancements:** Optimized Xplor Spot's dry cleaning management platform by building backend features to streamline order processing, payment integration, and customer interactions, significantly improving operational efficiency.
 - **Stakeholder-Driven Requirements Gathering:** Collaborated with stakeholders to define key system requirements, translating business objectives into technical specifications and implementing solutions aligned with user needs.
 - **End-to-End Testing & Debugging:** Developed automated and manual test cases to ensure feature reliability and robustness, conducting extensive debugging to maintain system scalability and performance.
- Software Engineer** Houston, TX
• *ACT@Rice — React Native, Node.js, WordPress* August 2023 - May 2024
 - **Mobile Application Development:** Led a team to design and develop a mobile app for Recipe for Success, a nonprofit focused on promoting healthy eating habits, incorporating user-centered design principles.
 - **Gamified Challenge Tracker:** Engineered a social tracking system with leaderboard functionality for users to log vegetable consumption, utilizing React Native's state management and backend connections.
 - **Scalable Architecture:** Constructed a scalable front-end architecture with React Native and integrated with WordPress backend via Node.js, ensuring seamless data flow and efficient user experience.
- Front-End Software Engineering Intern** Beaumont, TX
• *CRDN — JavaScript, HTML, CSS, Node.js* May 2023 - August 2023
 - **Backend for Restoration Services:** Developed core backend functionalities for CRDN's disaster recovery services website, enabling smooth request handling for textiles, art, and electronics restoration.
 - **Service Request Automation:** Implemented dynamic service request forms and backend logic to streamline customer service workflows, reducing processing time and optimizing user response during disaster recovery.
- Data Intern** Beaumont, TX
• *Sunshine Stores Inc. — SQL, Excel, Python* March 2022 - August 2022
 - **Operational Data Insights:** Conducted in-depth data analysis to provide insights on store performance, identifying trends that informed decision-making and optimized daily business operations.
 - **Data Migration & Quality Assurance:** Supported data migration efforts by performing data cleansing and validation, ensuring data accuracy and integrity for a seamless transition to a new management system.

PROJECTS

- **Employee Management System (Full Stack):** Developed a full-stack system for Munro's Cleaners to manage employee records, roles, and payroll, featuring real-time updates for streamlined administrative tasks. *JavaScript, CSS, HTML* (Aug '24)
- **OwlDB - NoSQL Document Database (Full Stack):** Designed and implemented OwlDB, a custom NoSQL database in Go, featuring document storage, retrieval, and a CLI for user interaction. *Golang* (Aug '24)
- **Bakery Website (Full Stack):** Created a responsive website for Batter Bar HTX to showcase dessert menus and contact details, enhancing user engagement and search engine visibility. *HTML, CSS, JavaScript* (May '24)
- **Hurricane Harvey Effect on Crime Rates (Data Science):** Analyzed crime rate shifts post-Hurricane Harvey using exploratory data analysis and machine learning, earning second place at Rice's DEEP competition. *Python, scikit-learn, numpy, pandas* (Dec '23)
- **Spot It Game (Projective Geometry):** Created an interactive "Spot It" card game based on projective geometry to enhance gameplay dynamics. *Python* (Dec '23)
- **Stock Prediction (Markov Chains):** Built a stock price prediction model using Markov Chains to analyze historical data and forecast price movements. *Python* (Oct '23)
- **Map Search Algorithms (Algorithm Design):** Implemented and compared three search algorithms—BFS, DFS, and A*—to optimize route finding on street maps, illustrating algorithmic efficiency. *Python* (July '23)

PROGRAMMING SKILLS

- **Languages:** Python, Java, JavaScript, CSS, HTML, C, Golang, Racket, TypeScript, SQL
Technologies: React, Pandas, Git, node.js, MySQL, Excel, PowerPoint, Word, LaTeX